

DTI INTEGRATION

Machine Learning Model for Cardiovascular Risk Prediction

TEAM-14

1.Team Name: CardioVision

Team Members:

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2. Problem/Opportunity Domain

Domain of Interest:

Healthcare – Preventive Cardiology and Artificial Intelligence

Focus Area:

Cognitive Load Estimation for Adaptive E-Learning Systems

Description:

Cardiovascular diseases (CVDs) are the leading cause of death worldwide, responsible for nearly 17.9 million deaths every year. Although risk prediction tools like the Framingham Risk Score exist, they rely only on limited clinical and lifestyle factors such as age, blood pressure, and cholesterol. These models often ignore environmental, stress, and socioeconomic factors that also influence heart health, leading to underestimation of risk.

This project aims to develop an **AI-powered system** that uses **machine learning algorithms** to predict the risk of cardiovascular disease using health records, lifestyle, and environmental data. It classifies patients as low, medium, or high risk and provides preventive recommendations through a user-friendly dashboard.

Why Chosen:

- Heart disease prevention is one of the most impactful areas where AI can save lives.
- Early prediction empowers people to make lifestyle changes before it's too late.
- This project combines healthcare and technology, contributing to both public health and innovation.

Aligns with Sustainable Development Goals (SDGs):

- SDG 3: Good Health and Well-being
- SDG 9: Industry, Innovation, and Infrastructure
- SDG 10: Reduced Inequalities

Alternatives & Limitations:

Existing Method	Limitations
Manual health checkups	Misses early warnings; costly and inconsistent
Doctor consultations based on symptoms	Reactive, not preventive
Traditional risk calculators	Limited variables, low adaptability, no continuous learning

Customers:

- General public (preventive health users)
- Healthcare professionals & doctors
- Fitness and wellness apps
- Insurance and health monitoring companies

Emotional Impact:

Many people feel anxious and helpless when unexpected cardiac issues occur despite an active lifestyle. A predictive system can build trust, reduce uncertainty, and empower users to take control of their health.

Quantifiable Impact:

- Up to **85% accurate risk prediction** using ML models
 - Early **detection** can reduce hospitalization by **20–25%**
 - Faster, **affordable** screening compared to lab tests
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Alternative Shortcomings:

- Lack of awareness and self-monitoring tools
- Medical consultations are expensive and time-consuming
- No real-time, personalized health feedback

3. Addressing SDGs (Sustainable Development Goals)

Relevant SDG	Contribution
SDG 3: Good Health and Well-being	Provides preventive healthcare and promotes lifestyle awareness.
SDG 9: Industry, Innovation, and Infrastructure	Integrates AI & IoT for modern medical innovation.
SDG 17: Partnerships for the Goals	Encourages collaboration among tech, healthcare, and research sectors.

4. Stakeholders

- Students / Learners
- Teachers / Educators
- E-Learning Platform Developers
- Educational Institutions
- Psychologists / Cognitive Scientists
- Parents and Mentors

Power–Interest Matrix of Stakeholders:

Category	Stakeholder Examples	Engagement Strategy
High Power, High Interest	Doctors, Hospitals	Involve in data validation & clinical interpretation
High Interest, Low Power	Users / Patients	Prioritize usability & awareness
Low Interest, High Power	Healthcare Authorities	Show policy-level impact
Low Power, Low Interest	General public (non-users)	Create awareness through campaigns

Empathetic Interviews

I Need to Know	Questions I Will Ask	Insights I Hope to Gain
Awareness of heart risk	How often do you get health checkups?	Understand preventive behavior
Lifestyle and stress	What habits do you think affect heart health?	Learn user mindset about health
Tech comfort	Would you trust an AI app for health predictions?	Assess digital health trust
Feedback preference	How should the system present results?	Build better UX and visualization

Skilled Interview Report

User/ Interviewee	Questions Asked	Insights Gained (NOT Their Answers)
Aashritha	How does the system predict heart disease risk?	Model uses features like cholesterol, glucose, BP, and BMI.
Indu	How is data collected securely?	All data anonymized and processed locally or via secure cloud.
Harshini	Can new users understand results easily?	Color-coded risk levels improve clarity.
Amulya	Can the model adapt to new users' lifestyles?	System can retrain periodically based on new patterns.
Pradeepa	How accurate is it?	Achieves ~85% accuracy using ensemble ML models.

7. Empathy Map

Who is your Customer?

Health-conscious individuals and patients at early risk.

Who are we empathizing with?

People who want to monitor their heart health easily without visiting hospitals frequently.

Key Points:

- Want quick, reliable results
 - Fear complex medical jargon
 - Seek personalized, friendly feedback
 - Desire privacy and data protection
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8. Persona of Stakeholders

Name: Ramesh, 35 years

Demographics: Working professional with mild hypertension

Goals: Stay fit, monitor health early

Challenges: Lack of time for regular checkups

Aspirations: Prevent future heart problems

Needs: Simple, AI-based health app

Pain Points: Confusion due to lack of medical awareness

9. Look for Common Themes, Behaviors, Needs, and Pain Points

- **Common Themes:** Awareness and early prevention
- **Behaviors:** Occasional checkups, high work stress, sedentary lifestyle
- **Needs:** Reliable predictions, health tips, easy interface
- **Pain Points:** Fear of disease, costly medical tests

10. Define Needs and Insights of Your Users

User Needs:

- Accurate, data-driven prediction
- Preventive lifestyle advice
- Easy-to-use, visually clear app
- Privacy and reliability

User Insights:

- People underestimate heart risks.
- They prefer visual results (colors, bars) over text.
- Motivation improves when advice is simple and actionable.

13. POV Statements

PoV Statement	PoV Question
Users need early awareness to prevent future heart disease.	How might we predict risk before symptoms appear?
Users fear medical terms and complexity.	How might we simplify results using visuals?
Users need motivation to maintain heart health.	How might we personalize lifestyle advice?
Users want to trust AI health predictions.	How might we ensure accuracy and transparency?

14. “How Might We” (HMW) Questions

User Need / Insight	HMW Question
Users need accurate results	How might we improve model precision through data diversity?
Users want simple feedback	How might we present predictions in clear, color-coded visuals?
Users need privacy	How might we ensure secure, anonymized data handling?
Users want motivation	How might we include adaptive health tips in the app?

16. Design Challenge Statement

Design Challenge:

Design a real-time, intelligent, and user-friendly AI system that predicts cardiovascular risk using health and lifestyle data, provides clear visual feedback, and offers personalized preventive recommendations to promote early intervention and heart wellness.

17. Validation Plan

Objective:

Ensure the Cardiovascular heart disease prediction is accurate, trustworthy, and user-friendly.

Stakeholder/User	Role	Feedback	Suggestions
Students	Test Users	Found predictions useful and easy to interpret	Add lifestyle-based recommendations
Doctors	Domain Experts	Appreciated clinical feature relevance	Improve medical explanation behind results
Developers	Technical Reviewers	Model performs efficiently	Optimize for deployment speed

18. Ideation

Idea No.	Proposed Solution	Key Features/Benefits	Challenges/Concerns
1	ML-based Heart Risk Prediction	Predicts CVD risk using patient data	Requires quality dataset
2	Interactive Dashboard	User-friendly Streamlit interface	UI clarity and trust
3	AI Health Advice Generator	Personalized tips for lifestyle improvement	Must ensure credible advice
4	Explainable AI Module	Displays factor influence on prediction	Visualization complexity
5	Secure Data Architecture	Protects user data & ensures privacy	Data encryption and policy

□ Solution Concept Form – Machine Learning Model for Cardiovascular Risk Prediction.

Problem Statement:

Heart diseases remain the leading cause of mortality worldwide. Many individuals are unaware of their cardiac risk until serious complications occur. Existing screening methods rely on traditional health checkups or limited calculators like the Framingham Risk Score, which fail to consider multiple influencing factors such as lifestyle, glucose levels, stress, and body mass index.

There is a need for an intelligent, AI-based system that can predict cardiovascular risk early using health and lifestyle data, enabling users to take preventive action before the onset of severe disease.

